

DANIIL DMITRIEV

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EDUCATION

PhD student, Department of Mathematics

ETH Zurich

ETH AI Center Fellow, advised by Prof. Afonso S. Bandeira and Prof. Fanny Yang

Thesis: *High-Dimensional Optimization Problems in Average-Case and Worst-Case Settings*

2021 to 2025

Master's of Data Science

EPFL

GPA: 5.57 out of 6.0

2018 to 2021

Bachelor's of Applied Mathematics and Computer Science

MIPT

GPA: 3.77 out of 4.0

2014 to 2018

RESEARCH EXPERIENCE

Postdoctoral Researcher, University of Pennsylvania

Philadelphia, Sep 2025 – present

Hosted by Prof. Yuting Wei in the Department of Statistics and Data Science at the Wharton.

PUBLICATIONS

DD*, Z. Huang*, Y. Wei, *Efficient sampling with discrete diffusion models: sharp and adaptive guarantees*, COLT, 2026
(* denotes equal contribution)

DD, H. Franck, C. Heinzler, A. Sanyal, *Learning in an echo chamber: online learning with replay adversary*, SODA, 2026
(alphabetical order)

A. Bandeira, J. Błasiok, **DD**, U. Faure, A. Kireeva, D. Kunisky, *The Lovász number of random circulant graphs*, SampTA, 2025 (alphabetical order)

DD*, R.-D. Buhai*, S. Tiegel, A. Wolters, G. Novikov, A. Sanyal, D. Steurer, F. Yang, *Robust mixture learning when outliers overwhelm small groups*, NeurIPS, 2024

DD, K. Szabó, A. Sanyal, *On the growth of mistakes in differentially private online learning: a lower bound perspective*, COLT, 2024

D. Schröder*, **DD***, H. Cui*, B. Loureiro, *Asymptotics of learning with deep structured (random) features*, ICML, 2024

G. Arpino, **DD**, N. Grometto, *Greedy heuristics and linear relaxations for the random hitting set problem*, APPROX, 2024
(alphabetical order)

D. Schröder, H. Cui, **DD**, B. Loureiro, *Deterministic equivalent and error universality of deep random features learning*, ICML, 2023

DD, M. Zhukovskii, *On monotonicity of Ramanujan function for binomial random variables*, Statistics & Probability Letters, 2021 (alphabetical order)

T. Lin, S. U. Stich, L. Barba, **DD**, M. Jaggi, *Dynamic model pruning with feedback*, ICLR, 2020

DD, M. Zhukovskii, *On a connection of two theoretical graph problems with conjectures of Ramanujan and Samuels*, Russian Mathematical Surveys, 2018 (alphabetical order)

TALKS

IMS Annual Meeting, Invited talk, 2026

University of Copenhagen, DeLTA seminar, 2024

ETH Zurich, DACO seminar, 2024

ETH Zurich, Graduate seminar in probability, 2023

ICTP, Trieste, Youth in High Dimensions, 2024

Les Diablerets, Workshop on spin glasses, 2022

INDUSTRY EXPERIENCE

Bloomberg L.P. Research Intern

London, Sep 2020 – Jan 2021

Applied diversification methods to improve the output of the recommendation system. Created pipeline for the experiments and compared the common diversification approaches (MMR, DPP) across multiple metrics.

Amazon Software Engineering Intern

Berlin, Jul 2018 – Sep 2018

Implemented image captioning in Python, MXNet. Optimized non-differentiable objectives for evaluating quality of the caption. Showed advantage compared to differentiable loss functions.

Google Software Engineering Intern

New York City, Jul 2017 – Oct 2017

Used clustering and time series analysis (DBSCAN, Granger Causality) in Python and Java to find dependencies in complex multivariate temporal data.

TEACHING

ETH Zurich (TA), Mathematics of Data Science (Fall 2021), Mathematics of Machine Learning (Spring 2022), Introduction to Machine Learning (Spring 2025)

EPFL (TA) Artificial Neural Networks (Spring 2020, Spring 2021)

Supervising MSc theses at ETH Zurich: Carolin Heinzler (Fall 2023), Krish Agrawal, Ulysse Faure (Spring 2024)

Teaching at Berkeley Math Circle (Fall 2024)

SERVICE

Reviewer at conferences: NeurIPS 2024 (Top reviewer), ICLR 2025, ICML 2025, COLT 2025, ICLR 2026, AISTATS 2026, COLT 2026

Reviewer at journals: TMLR, SIAM Journal on Mathematics of Data Science

Organizer of Theory breakfast at ETH Zurich (Spring 2024, 2025)